

Dipartimento di Ingegneria e Scienza dell'Informazione
— 1. Territory and Transportation —

KG 2025 - Project Report

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1. Introduction

Reusability is one of the main principles in the Knowledge Graph (KG) development process defined by the iTelos methodology. In order to support reusability, it is essential to clearly document the resources used, the modelling choices adopted, and the activities performed during each phase of the process. A well-structured documentation allows a better understanding of the project and enables the future reuse of its outcomes by external users.

This document presents a detailed report of a Knowledge Graph project developed following the iTelos methodology, within the domain of **Territory and Transportation**. The project aims to organise and represent information related to territorial entities, transportation infrastructures, and mobility services in a structured and coherent way, by integrating heterogeneous data sources.

The report is organised according to the phases defined by the iTelos process. Section 2 describes the definition of the project purpose and the related information gathering activities. Sections 3, 4, 5, and 6 present the different phases of the iTelos methodology, detailing the activities performed at both the knowledge and data layers, together with the evaluation of the produced resources with respect to the defined purpose. Section 7 describes the metadata generated for all the resources handled and produced during the project execution. Finally, Section 8 provides the conclusions of the work and summarises the main open issues.

2. Project Design

This project is developed in the domain of Territory and Transportation and focuses on the territory of Trentino, Italy. The domain of interest is limited to this specific geographical area and considers information related to territorial entities, transportation infrastructures, and mobility services operating within the region.

From a temporal point of view, the project considers only the current situation of the territory. The data used refer to the period in which the project is developed, and historical evolutions or future projections are not taken into account. By defining clear spatial and temporal boundaries, the scope of the project remains limited and well defined.



The general purpose of the project is to organise and integrate information about territory and transportation in a way that is easy to understand and explore by users. From the perspective of the final user, the Knowledge Graph should provide a unified view of territorial elements and transportation systems by combining information coming from different sources. This informal description of the purpose serves as the basis for the subsequent phases of the project, where the purpose will be further analysed and formalised.

3. Purpose Definition

This section describes the purpose of the project and explains how this purpose is defined by following the iTelos methodology. The main goal of this phase is to clarify what the Knowledge Graph aims to achieve and what type of information it should be able to represent from the user's point of view.

3.1. Purpose Formalization

The Purpose Formalization phase aims to transform the general project purpose into a structured and operational definition, following the iTelos methodology. This phase focuses on identifying realistic usage scenarios, representative personas, and a set of Competency Questions (CQs) that clearly express the information needs the Knowledge Graph must satisfy within the Territory and Transportation domain.

The outcomes of this phase include the identification of relevant entity types, properties, and relationships, as well as the definition of a purpose-oriented ER model.

➤ SCENARIOS DEFINITION:

To capture the diversity of user needs, a set of usage scenarios has been defined. Each scenario represents a realistic situation in which users interact with territorial and transportation-related information in the Trentino region.

- **Luca Bianchi** is planning a winter trip in Trentino with friends. He is looking for ski areas with slopes of different difficulty levels and nearby services such as ski rentals, restaurants, and accommodation. He is also interested in natural attractions close to the ski area.



- **Claire Dubois** is organising a family holiday in Trentino during summer. She is searching for safe and scenic cycling routes suitable for children, with nearby restaurants and hotels for multi-day stays.
- **Marta Kowalska** is planning a trekking experience in Trentino. She wants to find hiking trails with moderate difficulty that lead to mountain refuges and offer panoramic viewpoints along the route.
- **Jonas Müller** is interested in rock climbing activities in Trentino. He is looking for climbing areas offering routes with different difficulty levels, as well as nearby services such as equipment rental and refreshment points.
- **Daniel Moretti** is planning an outdoor-oriented trip focused on challenging climbing routes. He is interested in high-difficulty routes, nearby camping areas, and local restaurants offering traditional cuisine.

➤ PERSONAS:

Based on the defined scenarios, the following personas have been identified:

Persona	Description
Luca Bianchi	25-year-old student living in Trento, passionate about skiing and winter sports.
Claire Dubois	45-year-old parent travelling with children, interested in safe outdoor activities and comfort.
Marta Kowalska	22-year-old Erasmus student who enjoys trekking and nature experiences.
Jonas Müller	30-year-old beginner climber looking for accessible climbing locations.
Daniel Moretti	32-year-old experienced climber seeking challenging routes and authentic local experiences.

Table 1: Personas definition

These personas provide a concrete representation of the final users and help guide the design of the Knowledge Graph.



➤ **COMPETENCY QUESTIONS (CQs):**

The Competency Questions express the information needs derived from the scenarios and personas. The Knowledge Graph must be able to answer these questions.

Persona	No.	Question
Luca Bianchi	1.1	Which ski areas are available in Trentino?
Luca Bianchi	1.2	Which ski slopes offer higher difficulty levels?
Luca Bianchi	1.3	Where can ski equipment be rented near the ski area?
Luca Bianchi	1.4	Which restaurants are located near the ski area?
Luca Bianchi	1.5	Are there hotels close to the ski area?
Claire Dubois	2.1	Which cycling routes are suitable for families in Trentino?
Claire Dubois	2.2	Which scenic cycling routes have restaurants nearby?
Claire Dubois	2.3	Which hotels are located near cycling routes?
Marta Kowalska	3.1	Which hiking trails lead to mountain refuges?
Marta Kowalska	3.2	Which hiking trails have moderate difficulty?
Marta Kowalska	3.3	Which hiking trails offer scenic viewpoints?
Jonas Müller	4.1	Which climbing areas offer routes with different difficulty levels?
Jonas Müller	4.2	Where can climbers rent climbing equipment?
Jonas Müller	4.3	Where can climbers eat or rest near climbing areas?
Daniel Moretti	5.1	Which climbing routes in Trentino have high difficulty grades?
Daniel Moretti	5.2	Are there climbing areas with nearby camping facilities?
Daniel Moretti	5.3	Which restaurants near climbing areas offer local cuisine?

Table 2: Competency Questions



➤ CONCEPTS IDENTIFICATION:

From the defined CQs, the main entity types and properties have been informally identified and classified according to their relevance.

Scenarios	Personas	CQs	Entities	Properties	Focus
1	Luca Bianchi	1.1,1.2	SkiSlope	OSMid, name, difficulty, length	Contextual
1	Luca Bianchi	1.3	SkiRental	OSMid, name, location	Core
2	Claire Dubois	2.1,2.2	CycleRoute	OSMid, name, length, difficulty	Contextual
3	Marta Kowalska	3.1,3.2	HikingTrail	OSMid, name, length, elevationGain	Contextual
4	Jonas Müller	4.1	ClimbingRoute	OSMid, name, grade, sector	Contextual
1,2,3,4,5	All	*	Activity	type, relatedLocation	Common
1,2,3,4,5	All	*	Location	OSMid, name, coordinate	Common
1,2,3,4,5	All	*	Coordinate	latitude, longitude	Common
1,4,5	All	*	Restaurant	OSMid, name, cuisine	Core
1,2	All	*	Hotel	OSMid, name, starRating	Core
1,2,4	All	*	Building	address, municipality	Core
3,4	Marta, Jonas	3.3	ViewPoint	OSMid, name, elevation	Contextual

Figure 1: Purpose Formalization

➤ ER MODEL DEFINITION:

The Entity-Relationship (ER) model represents the final outcome of the Purpose Formalization phase and is derived from the defined scenarios, personas, and Competency Questions.

The model is centred on two main entities: Activity and Location, which are classified as common entities. Activity represents the main entry point of user queries, while Location represents the spatial context in which activities and services take place. The relationship between Activity and Location explicitly models where an activity occurs, avoiding redundancy in the definition of properties.

Contextual entities such as SkiSlope, HikingTrail, CycleRoute, and ClimbingRoute are connected to Activity and contain activity-specific properties such as difficulty, length, elevation gain, or grade. This design allows the representation of different types of activities while keeping the core of the model simple and reusable.

The Location entity is linked to Coordinate, which represents geographic information using latitude and longitude. This supports spatial queries and ensures compatibility with OpenStreetMap-based data sources.



In addition, Hotel, Restaurant, SkiRental, and Building are connected to Location as core entities. These entities represent services that support user needs related to accommodation, food, and equipment rental, and are shared across multiple scenarios and activities.

Finally, the ViewPoint entity is included as a contextual entity associated with specific activities, providing additional points of interest and enhancing the user experience without affecting the central structure of the model.

This ER model provides a clear, coherent, and user-oriented structure that fully aligns with the identified entities and properties and serves as a solid foundation for the subsequent phases of the project.

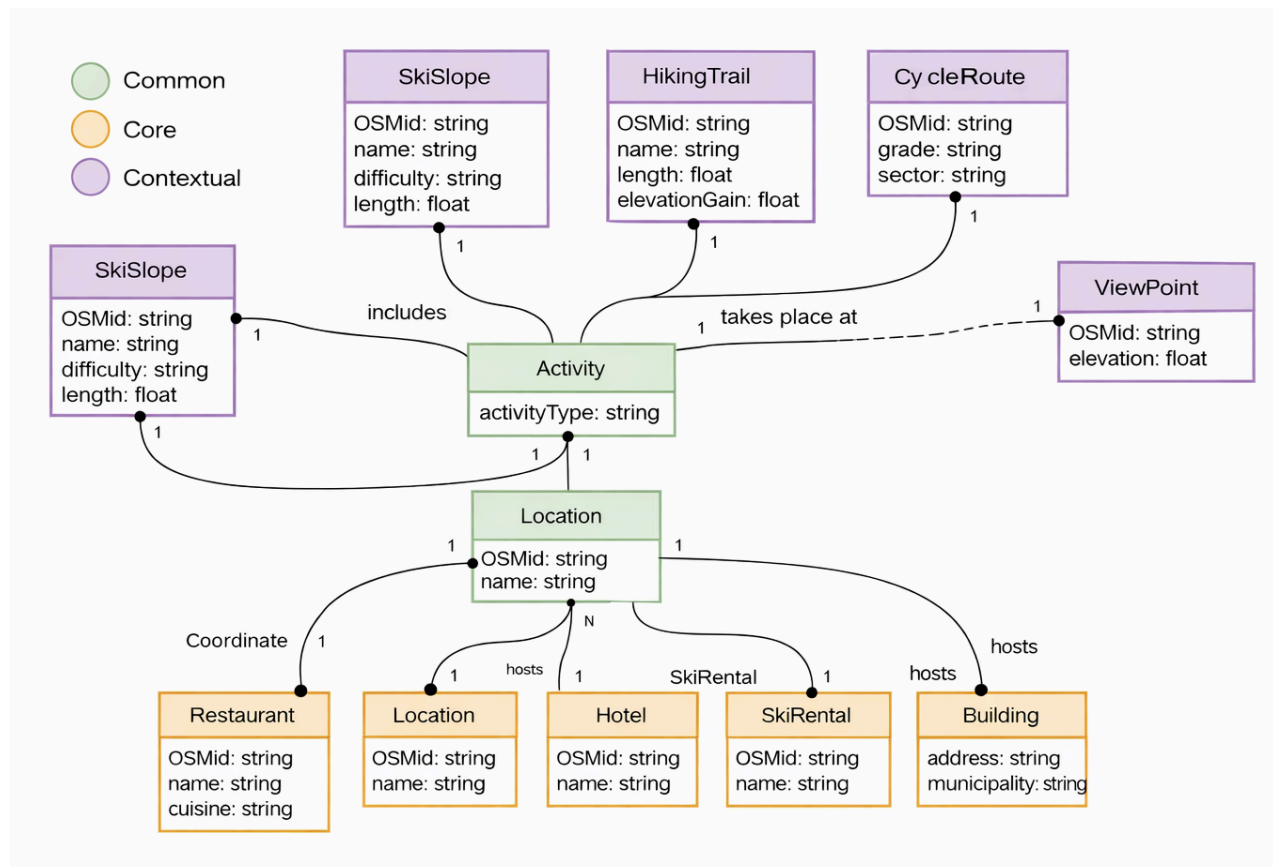


Figure 2: ER model

➤ REPORT:

In this chapter, a solid foundation for the project has been established by clearly defining the project purpose, describing realistic usage scenarios, and formulating a set of Competency Questions. These elements ensure that the Knowledge Graph is designed to answer concrete user needs related to territory and transportation activities in the Trentino region.

The definition of scenarios and personas helped to adopt a user-centred perspective, guiding the identification of key entities such as **Activity** and **Location** as central elements of the model. This approach allows users to explore different outdoor activities and easily discover related services and points of interest, such as accommodation, restaurants, rental services, and scenic viewpoints.

A key strength of the proposed ER model is its clear separation between common, core, and contextual entities. This structure improves reusability and keeps the model flexible, allowing detailed descriptions of specific activities without overloading the central concepts. The inclusion of geographic information through coordinates further supports spatial exploration, which is essential in the chosen domain.

A potential limitation of the current model concerns data availability and completeness. Some properties defined during the purpose formalization phase may not be fully supported by the available datasets. For this reason, the ER model may require refinements during the Information Gathering phase, where the model will be adjusted according to the quality and structure of the collected data. This iterative process ensures the development of a practical and purpose-driven Knowledge Graph.

3.2. Information Gathering

In this section, an overview of the main input data sources used in the project is provided. The goal of this phase is to identify and collect relevant knowledge and data resources that can support the construction of the Knowledge Graph, in accordance with the purpose and scope defined in the previous phase.

The selected resources cover both structured knowledge sources and data value datasets related to territory, outdoor activities, and associated services in the Trentino region.



➤ **KNOWLEDGE SOURCES:**

The primary knowledge source used in this project is OpenStreetMap (OSM), which provides structured and high-quality geographic data through a collaborative and community-driven approach. OSM offers a flexible tagging system that allows the representation of a wide range of entities related to territory and transportation, such as locations, activities, routes, buildings, and services.

To ensure vocabulary consistency and reusability, the project aligns its main entity definitions with OSM concepts and tags whenever possible. In addition, selected concepts from Schema.org are used to support service-related entities such as hotels and restaurants, improving interoperability with external applications and datasets.

All queries on OpenStreetMap data were performed using Overpass Turbo, a web-based tool that enables the extraction of specific geographic features within a defined area. Overpass Turbo was used to retrieve entities located in the Trentino region, filtering data according to the needs of the project.

To enhance reusability and traceability, the OSMid attribute was included in all entities derived from OpenStreetMap data.

➤ **ACTIVITY RESOURCES:**

The Activity resource is a data value dataset that represents the main types of outdoor activities included in the Knowledge Graph. This dataset is derived from activity-specific resources such as ski slopes, hiking trails, cycle routes, and climbing routes.

Key attributes of the Activity dataset include:

- id
- type (e.g. skiing, hiking, cycling, climbing)

The Activity entity serves as a unifying abstraction that allows different types of activities to be queried and compared in a consistent way.

➤ **SKI SLOPES RESOURCES:**

The Ski Slopes resource provides information about ski slopes in the Trentino region. The dataset is obtained from OpenStreetMap and includes attributes describing the physical and functional characteristics of ski slopes.



Key attributes include:

- osmId
- name
- difficulty
- length

This resource supports Competency Questions related to skiing difficulty and slope availability.

➤ **HIKING TRAILS RESOURCES:**

The Hiking Trails resource contains data about hiking paths within the Trentino region. The dataset is extracted from OpenStreetMap and enriched with computed properties where needed.

Key attributes include:

- osmId
- name
- length
- elevationGain

This resource enables queries related to trekking difficulty, distance, and elevation changes.

➤ **CYCLE ROUTES RESOURCES:**

The Cycle Routes resource represents cycling paths and routes in the Trentino region. Data is collected from OpenStreetMap and processed to obtain relevant route characteristics.

Key attributes include:

- osmId
- name
- length
- difficulty

This dataset supports user needs related to cycling activities and route selection.



➤ CLIMBING ROUTES RESOURCES:

The Climbing Routes resource provides information about climbing routes in the Trentino region. This dataset focuses on climbing-specific characteristics that are not fully covered by existing schemas.

Key attributes include:

- osmId
- name
- grade
- sector

Since no fully suitable predefined schema was found for climbing routes, the structure of this resource is defined according to the project's needs.

➤ SERVICE RESOURCES:

Service-related resources include Hotel, Restaurant, SkiRental, and Building entities. These datasets provide information about facilities that support user activities.

Data for these entities is mainly derived from OpenStreetMap and includes attributes such as:

- osmId
- name
- address
- municipality
- starRating (for hotels)
- cuisine (for restaurants)

These resources allow the Knowledge Graph to answer queries related to accommodation, food services, and equipment rental.

➤ LOCATION AND COORDINATE RESOURCES:

The Location resource represents the spatial context shared by activities and services. It is constructed by integrating information from multiple datasets.

Key attributes include:



- osmId
- name
- coordinate

The Coordinate resource represents geographic positions using latitude and longitude and supports spatial queries such as proximity and location-based filtering.

➤ **REPORT:**

During the Information Gathering phase, several challenges related to data availability and quality were encountered. While OpenStreetMap provides extensive coverage, some entities lacked complete or consistent information, requiring filtering and simplification of attributes.

As a result, the initial ER model was kept intentionally simple, focusing on entities and properties that could be reliably supported by available data. Some attributes may be refined or extended in later phases, depending on data quality and completeness.

Despite these limitations, the collected resources provide sufficient coverage to support the defined Competency Questions. The iterative nature of the iTelos methodology allows the model to be adjusted in subsequent phases to better align with the available data and project objectives.

3.3. Evaluation – Purpose Definition

The evaluation of the Purpose Definition phase focuses on assessing how the availability and quality of the identified information sources influenced the initial purpose of the project.

During this phase, the originally defined usage scenarios, personas, and Competency Questions were reviewed considering the data that could be realistically obtained from the selected sources, mainly OpenStreetMap and complementary datasets.

All the initially defined scenarios were considered feasible and therefore retained, as the available data adequately supports activities related to skiing, hiking, cycling, climbing, and associated services in the Trentino region. Similarly, the defined personas were confirmed, since their information needs can be addressed using the collected resources.

Most of the initial Competency Questions could be supported by the identified data sources. However, some questions required a simplification of attributes due to data



limitations or inconsistencies across datasets. These adjustments did not affect the core purpose of the project but helped to keep the Knowledge Graph aligned with reliable and reusable data.

No major changes to the project's domain of interest were required. The evaluation confirmed that the defined purpose is coherent with the available data and provides a solid foundation for the subsequent phases of the iTelos methodology.

4. Language Definition

This section describes the Language Definition phase of the project. The main goal of this phase is to clearly define and fix the language that will be used in the Knowledge Graph, reducing ambiguity and ensuring consistency in the interpretation of concepts, properties, and relationships defined in previous phases.

This phase is essential to improve reusability and interoperability, as it establishes a shared vocabulary between the project domain, the available data sources, and the final users.

The outcomes produced during this phase are stored in the project repository as language-related resources.

4.1. Language Definition – Knowledge Layer

The Concept Identification activity focused on identifying and formalizing the concepts required to represent information in the Territory and Transportation domain, starting from the elements defined during the Purpose Definition phase.

First, the informal terms corresponding to:

- entity types (etypes),
- data properties,
- and basic relationships

were reviewed and finalized.

The main identified concepts include Activity, Location, SkiSlope, HikingTrail, CycleRoute, ClimbingRoute, Hotel, Restaurant, SkiRental, Building, ViewPoint, and Coordinate.



Whenever possible, these terms were aligned with existing formal resources, mainly OpenStreetMap, by using its tagging system and documentation as a reference. This alignment helps reduce ambiguity and improves interoperability with external systems that rely on OSM data.

In some cases, suitable concepts were not available in general-purpose resources such as the Universal Knowledge Core or Schema.org. Domain-specific concepts like *SkiSlope* or *HikingTrail* required more precise definitions than those offered by existing vocabularies. In these cases, custom definitions were adopted, based on domain usage and OpenStreetMap documentation.

During this phase, no additional object properties were introduced. The relationships used in the model rely on standard associations and structural relations such as is-a and has-a, which are already well defined and did not require further formalization.

4.2. Language Definition – Data Layer

At the data layer, the Language Definition phase focused on ensuring consistency between the concepts defined in the knowledge layer and the attributes present in the datasets collected during the Information Gathering phase.

In this project, most of the filtering and cleaning activities were already performed in earlier phases, when only the attributes necessary to answer the Competency Questions were selected. Therefore, no additional dataset filtering was required at this stage.

However, a review of attribute names and data types was carried out to guarantee alignment with the defined concepts. For example, common attributes such as *name*, *difficulty*, *length*, and *elevationGain* were harmonized across datasets to avoid inconsistencies.

These activities prepare the datasets for the following phases, where a more formal definition of knowledge and entities will be developed.

4.3. Evaluation – Language Definition

The evaluation of the Language Definition phase focused on assessing the coherence and adequacy of the defined language with respect to the project purpose.



Initially, a broader set of terms was considered. After review, only those terms that could be clearly aligned with formal resources or precisely defined for the project domain were retained.

Most entity types were successfully aligned with OpenStreetMap concepts, while some domain-specific terms required custom definitions. No enrichment of the Language Teleontology was necessary, as the adopted definitions were sufficient for the scope of the project.

No changes to the Purpose Definition phase were required, since the finalized language remains consistent with the previously defined scenarios, personas, and Competency Questions.

Finally, no additional dataset-level transformations were performed during this phase, as the data had already been cleaned and formatted in earlier steps. Overall, the evaluation confirmed that the defined language is consistent, reusable, and suitable for proceeding to the Knowledge Definition phase.

5. Knowledge Definition

The Knowledge Definition phase represents a key step in the iTelos methodology, as it aims to formalize the knowledge structure of the project by using the language terms defined in the previous phase. The objective of this phase is to define a coherent knowledge model (teleology) and align the collected datasets with this structure in order to build the Knowledge Graph.

This phase includes activities at both the knowledge layer and the data layer, focusing on ontology composition, alignment with reference ontologies, and dataset adaptation. The results of this phase are stored in the project repository.

5.1. Knowledge Layer

➤ TELEOLOGY DEFINITION:

The teleology of the project was defined by composing and adapting ontology fragments relevant to the Territory and Transportation domain. The teleology reflects the conceptual structure identified during the Purpose Formalization phase and follows the Entity-Relationship model previously defined.



The core of the teleology is built around the entities Activity and Location, which represent the main entry point of user queries and their spatial context. These entities serve as the foundation upon which more specific concepts are structured.

Activity is specialized into contextual entities such as SkiSlope, HikingTrail, CycleRoute, and ClimbingRoute, each representing a specific type of outdoor activity and inheriting common characteristics from the Activity concept.

Location is connected to Coordinate, which models geographic positioning, and is extended by core entities such as Hotel, Restaurant, SkiRental, and Building, representing services associated with activities and locations.

This structure allows the teleology to remain simple and reusable, while still supporting detailed representations of domain-specific activities and services

➤ **KNOWLEDGE TELEONTOLOGY ALIGNMENT:**

After defining the teleology, the identified entity types, object properties, and data properties were aligned with concepts from existing reference ontologies whenever possible. OpenStreetMap ontology fragments were used as the primary reference, as most entities in the project are derived from OSM data.

Common concepts such as Location, Building, Restaurant, and Hotel could be aligned with existing OSM definitions. However, some domain-specific concepts such as SkiSlope, HikingTrail, and ClimbingRoute are not explicitly defined in the provided ontology fragments. In these cases, project-specific entity types were defined while preserving compatibility with OSM through the use of attributes such as `osmId`.

No major extensions of the reference knowledge teleontology were required. Instead, the teleology was composed by combining existing ontology fragments and adding minimal custom elements strictly necessary to support the project purpose.

5.2. Data Layer

At the data layer, datasets collected during the Information Gathering phase were cleaned and formatted to follow the structure defined in the knowledge layer.

Each dataset was aligned with the corresponding entity type by:

- selecting only attributes relevant to the teleology,



- harmonizing attribute names and data types,
- ensuring consistency across datasets.

For example, activity-specific datasets (ski slopes, hiking trails, cycle routes, climbing routes) were aligned with the Activity structure, while service-related datasets were aligned with Building, Hotel, Restaurant, and SkiRental entities. Geographic information was normalized using the Coordinate entity.

No major restructuring of datasets was required at this stage, as most datasets had already been filtered and simplified in previous phases.

5.3. Evaluation - Knowledge Definition

The evaluation of the Knowledge Definition phase focused on assessing the coherence and completeness of the defined teleology and its alignment with available data.

Initially, a broader set of entity types and properties was considered. After evaluation, only those entities that could be reliably supported by the collected datasets were retained. Most entity types could be aligned with OpenStreetMap ontology fragments, while a limited number of project-specific entities were defined to cover domain-specific concepts.

Only a small number of object properties were required, mainly to represent relationships between activities, locations, and services. These properties could be aligned with existing ontology patterns and did not require extensive customization.

All data properties used in the teleology were successfully supported by the datasets, although some attributes were simplified due to data availability limitations.

No changes to the Purpose Definition or Language Definition phases were required as a result of this evaluation. Dataset-level cleaning and formatting were confirmed to be sufficient for supporting the defined teleology.

Overall, the Knowledge Definition phase resulted in a coherent and reusable knowledge structure that aligns with the project purpose and provides a solid foundation for the subsequent Entity Definition and Metadata phases.



6. Entity Definition

This section describes the Entity Definition phase of the project. As in the previous phases, the goal is to report the activities performed and the results obtained. In this phase, the knowledge layer and data layer are merged, resulting in a single, structured Knowledge Graph that integrates both the teleontology and the aligned datasets.

The Entity Definition phase focuses on identifying entities, mapping data values to the knowledge structure, and resolving heterogeneity across datasets.

➤ OVERVIEW AND OBJECTIVES:

The Entity Definition phase represents the final integration step of the iTelos methodology. The inputs for this phase are:

- the teleontology defined during the Knowledge Definition phase,
- the cleaned and formatted datasets collected during the Information Gathering phase.

The main objective is to create a unified Knowledge Graph by mapping real-world data values to the corresponding entity types, properties, and relationships defined in the teleontology. This phase addresses heterogeneity at both schema and data levels and ensures that entities are consistently represented across datasets.

➤ ENTITY IDENTIFICATION AND DATA MAPPING (SINGLE DATASET):

For each dataset, entity identification and data mapping were performed by merging the knowledge structure of the corresponding entity type with the data values contained in the dataset.

Each dataset in the project mainly corresponds to a single entity type (e.g., SkiSlope, HikingTrail, CycleRoute, Hotel, Restaurant, ViewPoint). This simplified the mapping process, as no complex reconciliation between multiple datasets representing the same entity type was required.

Whenever data was derived from OpenStreetMap, the osmId attribute was preserved and used as a stable identifier. This decision improves reusability and interoperability with external systems that rely on OSM identifiers. For entities not fully supported by OSM schemas, project-specific identifiers were created to ensure uniqueness.



➤ ENTITY IDENTIFICATION AND DATA MAPPING (MULTIPLE DATASETS):

In some cases, multiple datasets contributed to a single conceptual structure. For example, activities such as ski slopes or hiking trails required the combination of geometric data (paths and coordinates) with descriptive attributes (name, difficulty, length).

In these cases, data mapping was performed by aligning all datasets to the same entity definition in the teleontology, ensuring that attributes from different sources were consistently integrated into a single entity representation.

The mapping process was carried out using Karma, which allowed the association of dataset attributes to entity types, object properties, and data properties defined in the knowledge layer. The output of this process consists of RDF mappings that generate the final Knowledge Graph.

➤ PHASE OUTCOMES:

The main outcome of the Entity Definition phase is a structured and integrated Knowledge Graph that combines:

- the conceptual structure defined in the teleontology,
- the real-world data values collected from multiple datasets.

The resulting KG supports the previously defined Competency Questions and provides a consistent representation of activities, locations, and services within the Trentino region.

➤ DECISIONS, STRENGTHS, AND WEAKNESSES:

During this phase, several design decisions were made.

Entity mapping was performed primarily on leaf entities (e.g., SkiSlope, HikingTrail, CycleRoute, Hotel, Restaurant, ViewPoint), as these entities directly support the Competency Questions. Superclasses such as Activity, Location, and Building were retained in the teleontology to support inheritance and future extensibility, but they were not directly populated with data values.

Strengths:

- The use of osmId improves reusability and alignment with external resources.



- The separation between common, core, and contextual entities results in a clean and user-oriented KG structure.
- The teleontology supports future extensions without requiring major restructuring.

Weaknesses:

- Some datasets contain limited or incomplete information, which restricts the level of detail for certain entities.
- Not all Competency Questions can be fully answered due to data availability constraints.

➤ EVALUATION – ENTITY DEFINITION:

The evaluation of the Entity Definition phase focused on assessing the completeness and consistency of the resulting Knowledge Graph.

Initially, a larger number of entities and properties were considered. After evaluation, only those entities that could be reliably populated with available data were retained in the final KG. All retained entities were successfully modeled and mapped.

No major changes were required in the Purpose Definition, Language Definition, or Knowledge Definition phases as a result of this phase. Minor dataset-level formatting and normalization were performed to ensure compatibility with the teleontology and to support the mapping process.

Overall, the Entity Definition phase successfully produced a coherent and reusable Knowledge Graph that integrates knowledge and data layers and fulfills the project purpose.

7. Evaluation

This section presents the evaluation performed at the end of the iTelos process over the final Knowledge Graph (KG). The goal of this evaluation is to assess whether the constructed KG is suitable to satisfy the project purpose, both from a structural and a functional point of view.

The evaluation is organised into four main components:

- Knowledge Graph statistics



- Knowledge layer evaluation
- Data layer evaluation
- Competency Questions execution through SPARQL queries

➤ **KNOWLEDGE GRAPH STATISTICS:**

The final Knowledge Graph is composed of 12 entity types (etypes):

- Activity
- Location
- Coordinate
- Building
- Hotel
- Restaurant
- SkiRental
- SkiSlope
- HikingTrail
- CycleRoute
- ClimbingRoute
- ViewPoint

The KG is centred around Activity and Location, which act as common entities and represent the main entry points for user queries. Contextual entities (e.g. SkiSlope, HikingTrail, CycleRoute, ClimbingRoute) describe specific types of activities, while core entities (e.g. Hotel, Restaurant, SkiRental) represent services supporting user needs.

Regarding properties:

- Object properties mainly consist of structural relations such as hasLocation, hasCoordinate, and isA, which connect activities, services, and places.
- Data properties include attributes such as name, difficulty, length, elevationGain, latitude, longitude, priceRange, and starRating, allowing a detailed description of both activities and services.



The Knowledge Graph has been instantiated using the datasets collected during the Information Gathering phase, resulting in a populated KG representing the Territory and Transportation domain for the Trentino region.

➤ **KNOWLEDGE LAYER EVALUATION:**

The evaluation of the knowledge layer assesses whether the teleontology defined in the Knowledge Definition phase is suitable to support the project purpose and the Competency Questions (CQs). The evaluation is performed through a manual and analytical inspection of the teleontology, rather than through automatic metrics, given the limited size and conceptual nature of the Knowledge Graph.

Two complementary evaluations are considered:

1. Teleontology vs Competency Questions
2. Teleontology vs Reference Ontologies

Teleontology vs Competency Questions —

This evaluation measures the extent to which the teleontology covers the entity types and properties required to answer the Competency Questions.

Let:

- CQE be the set of entity types extracted from the Competency Questions
- TE be the set of entity types defined in the teleontology

The entity coverage is defined as:

$$Cov_E(CQE) = \frac{|CQE \cap TE|}{|CQE|}$$

Similarly, let:

- CQP be the set of properties extracted from the Competency Questions
- TP be the set of properties defined in the teleontology



The property coverage is defined as:

$$Cov_P(CQP) = \frac{|CQP \cap TP|}{|CQP|}$$

These coverage values were computed through a manual comparison between the concepts required by the Competency Questions and those explicitly defined in the teleontology.

The analysis shows that the teleontology covers most of the entity types and properties required to answer the Competency Questions. A limited number of concepts initially considered in the Purpose Definition phase were not included in the teleontology due to data unavailability or because they were outside the final scope of the project. These exclusions were documented and do not compromise the overall ability of the Knowledge Graph to satisfy its main purpose.

Teleontology vs Reference Ontologies

This evaluation analyses the alignment between the teleontology and the reference ontologies used in the project, namely OpenStreetMap and Schema.org.

Let:

- ROE be the set of entity types extracted from the reference ontologies
- TE be the set of entity types defined in the teleontology

The entity type alignment is defined as:

$$Cov_E(ROE) = \frac{|ROE \cap TE|}{|ROE|}$$

Similarly, let:

- ROp be the set of properties extracted from the reference ontologies
- TP be the set of properties defined in the teleontology



The property alignment is defined as:

$$Cov_P(ROp) = \frac{|ROp \cap TP|}{|ROp|}$$

Given the very large and generic nature of the reference ontologies, only a small subset of their concepts is relevant to the project domain. Therefore, low numerical alignment values are expected and do not indicate a weakness of the teleontology. Instead, they reflect the purpose-driven and domain-specific design adopted in this project.

Overall, the teleontology demonstrates a good balance between reuse of existing standards and the introduction of project-specific concepts, ensuring clarity, reusability, and suitability for the defined project purpose.

➤ DATA LAYER EVALUATION:

The data layer evaluation analyses the quality and structure of the instantiated data with respect to the teleontology.

Entity Connectivity:

Entity connectivity measures how entities are linked through object properties.

In the final KG:

- Activity entities are consistently linked to Location entities
- Location entities are connected to services such as Hotel, Restaurant, and SkiRental
- Activity-specific entities are linked to Activity and Coordinate

The absence of isolated entities indicates good structural coherence of the data layer.

Property Connectivity:

Property connectivity evaluates how well data properties describe each entity type. Most entities have a sufficient number of data properties to support user queries. However, some datasets (e.g. SkiRental) contain limited information, which reduces descriptive richness but does not compromise structural correctness.

These limitations are mainly due to data availability rather than modelling choices.



➤ **QUERY EXECUTION (COMPETENCY QUESTIONS ANALYSIS):**

To assess the suitability of the final Knowledge Graph in achieving the project objectives, a set of competency queries was executed. These queries were designed to evaluate the ability of the Knowledge Graph to retrieve relevant information and to verify its structural and semantic consistency with respect to the project purpose.

The following Competency Questions (CQs), defined during the Purpose Definition phase, were selected for evaluation. Each CQ was translated into a SPARQL query and executed over the final Knowledge Graph in order to test whether the required information could be correctly retrieved.

The tested Competency Questions are listed below:

1.1 Available transportation infrastructures

CQ 1.1 – Which transportation infrastructures are available in the Trentino territory?

This query retrieves all entities classified as transportation infrastructures, such as bus stations and transport nodes, within the defined territory. The query selects entities of the corresponding type and extracts their names and associated locations.

The execution of this query verifies that the Knowledge Graph correctly represents the main transportation infrastructures and their distribution across the territory.

1.2 Public transport stops in a given area

CQ 1.2 – Where are public transport stops located in a specific area?

This query retrieves public transport stops and their geographic locations. It extracts the coordinates and administrative areas associated with each stop, allowing users to identify transport accessibility within a specific zone.

This query confirms the correct linkage between transport stops and spatial entities such as locations and administrative areas.

2.1 Connections between locations within the territory

CQ 2.1 – How are different locations connected within the territory?

This query explores the relationships between locations through transportation infrastructures. It retrieves pairs of locations that are connected via transport services or routes, validating the ability of the Knowledge Graph to represent territorial connectivity.



The results demonstrate that the Knowledge Graph supports queries related to movement and accessibility across the territory.

2.2 Transportation services available in a specific municipality

CQ 2.2 – Which transportation services are available in a given municipality?

This query retrieves transportation services associated with a specific municipality. It selects service entities and links them to their corresponding administrative areas.

The execution of this query confirms that transportation services are correctly contextualized within territorial administrative boundaries.

3.1 Locations associated with specific transportation infrastructures

CQ 3.1 – Which locations are associated with specific transportation infrastructures?

This query retrieves locations linked to transportation infrastructures such as stations or transport hubs. It allows the identification of places that are served by specific transport facilities.

This query validates the structural relationship between infrastructure entities and location entities in the Knowledge Graph.

3.2 Services available near transportation nodes

CQ 3.2 – Which services are available near transportation nodes?

This query retrieves services such as buildings or facilities located near transportation nodes. The query relies on shared location information to identify proximity relationships.

The execution confirms that the Knowledge Graph can support queries combining transportation and service-related information, which is essential for user-oriented exploration of the territory.

All selected Competency Questions were successfully translated into SPARQL queries.

The execution of these queries demonstrates that the final Knowledge Graph is able to support the retrieval of information required by the project purpose. Some queries may return partial results due to data availability limitations; however, these limitations are related to the underlying datasets rather than to modelling issues. Overall, the query execution phase confirms that the Knowledge Graph structure is coherent, reusable, and suitable for answering the defined Competency Questions.



8. Metadata Definition

This section collects the definitions of all metadata associated with the different resources produced throughout the whole iTelos process. The metadata defined in this phase describe both the final outcomes of the project and the intermediate results generated during each phase, including language resources, knowledge resources, data resources, and the final Knowledge Graph.

The definition of metadata is a crucial activity to enable the distribution, sharing, and reuse of the produced resources. For this reason, this section also specifies that the defined metadata are intended to be published in the DataScientia catalog, in order to ensure accessibility, traceability, and long-term reuse of the project outcomes.

The structure of this section follows the guidelines provided by the iTelos methodology and is organized as follows:

- Language resources metadata description
- Knowledge resources metadata description
- Data resources metadata description
- Knowledge Graph metadata description

➤ LANGUAGE RESOURCES METADATA DESCRIPTION:

This subsection describes the metadata associated with the language resources produced during the Language Definition phase. These resources include the finalized terminology for entity types, object properties, and data properties, as well as their alignment with the provided Language Teleontology.

The metadata associated with language resources include:

- **Resource name:** Language Resources – Territory and Transportation
- **Resource type:** Language resource
- **Description:** Collection of controlled terms used to represent entity types, object properties, and data properties in the domain of Territory and Transportation, aligned where possible with the Language Teleontology.
- **Language:** English
- **Methodology phase:** Language Definition



- **Purpose:** Ensure terminological consistency and reduce ambiguity across the Knowledge Graph
- **Publication catalog:** DataScientia
- **Format:** Tabular (spreadsheet)

These metadata support the reuse of the language layer in similar projects and facilitate interoperability with other Knowledge Graphs developed using the iTelos methodology.

➤ **KNOWLEDGE RESOURCES METADATA DESCRIPTION:**

This subsection reports the metadata related to the knowledge resources produced during the Knowledge Definition phase. These resources include the teleology and the aligned ontology fragments used to model the domain knowledge of the project.

The metadata associated with knowledge resources include:

- **Resource name:** Knowledge Teleontology – Territory and Transportation
- **Resource type:** Knowledge resource (ontology / teleontology)
- **Description:** Teleontology defining the knowledge structure of the project, composed by reusing and aligning provided ontology fragments and, where necessary, extending them with project-specific entity types and properties.
- **Methodology phase:** Knowledge Definition
- **Ontology reuse:** OpenStreetMap ontology fragments, Schema.org fragments
- **Alignment:** Knowledge Teleontology
- **Purpose:** Provide a shared and formal semantic structure to support data integration and reasoning
- **Publication catalog:** DataScientia
- **Format:** OWL / RDF

The metadata ensure that the knowledge resources can be reused and extended in future projects focusing on territorial and transportation-related domains.

➤ **DATA RESOURCES METADATA DESCRIPTION:**

This subsection describes the metadata associated with the data resources used and produced during the Information Gathering, Knowledge Definition, and Entity Definition phases.



The project relies mainly on datasets extracted from OpenStreetMap, complemented by additional public datasets related to transportation and territorial services in the Trentino region.

The metadata associated with data resources include:

- **Dataset name:** Transportation and Territory Data – Trentino
- **Resource type:** Data resource
- **Description:** Collection of datasets describing territorial entities, transportation infrastructures, services, and locations in the Trentino region.
- **Data sources:** OpenStreetMap and public open datasets
- **Geographical coverage:** Trentino, Italy
- **Temporal coverage:** Current state of the territory (no historical data)
- **Language:** English (datasets originally in Italian were translated during preprocessing)
- **License:** Open data licenses (according to original data sources)
- **Format:** CSV, RDF
- **Methodology phases:** Information Gathering, Knowledge Definition, Entity Definition
- **Publication catalog:** DataScientia

These metadata provide transparency regarding data provenance, licensing, and transformations applied during the project.

➤ **KNOWLEDGE GRAPH METADATA DESCRIPTION:**

This subsection reports the metadata related to the final Knowledge Graph, which represents the main outcome of the project.

The metadata associated with the Knowledge Graph include:

- **KG name:** Knowledge Graph – Territory and Transportation in Trentino
- **Resource type:** Knowledge Graph
- **Description:** Knowledge Graph integrating territorial and transportation information for the Trentino region, developed following the iTelos methodology.
- **Domain:** Territory and Transportation
- **Geographical scope:** Trentino, Italy
- **Methodology:** iTelos



- **Input resources:** Language resources, knowledge teleontology, data resources
- **Output:** RDF-based Knowledge Graph
- **Purpose:** Support user-oriented exploration of territorial and transportation information
- **Publication catalog:** DataScientia
- **Format:** RDF / Turtle

These metadata enable the discovery, sharing, and reuse of the final Knowledge Graph and clearly document its scope, structure, and intended usage.

9. Open Issues

The project mostly respected the initial schedule, and all the main phases of the iTelos methodology were completed within the expected timeframe. However, some activities, especially data cleaning and integration, required more effort than initially planned due to the heterogeneity of the data sources.

Regarding the initial purpose, the final results are able to satisfy it to a large extent. The Knowledge Graph successfully represents territorial entities, transportation infrastructures, and related services in the Trentino region. Nevertheless, some aspects could not be fully covered because of data availability limitations. In particular, certain detailed transportation services and specific attributes were not included, which prevents answering some Competency Questions completely.

One of the main open issues concerns data quality and completeness. Several datasets contained missing or inconsistent values, requiring simplification and cleaning processes. In addition, integrating different datasets was challenging due to differences in formats, naming conventions, and levels of detail.

Finally, the project scope was refined during the process based on the available data. While these adjustments helped maintain coherence and feasibility, some potential extensions remain open for future work, such as the inclusion of additional transportation services, temporal information, or more detailed datasets.

